

Diophantine Equations & Integer Restrictions

Student Assimilation Pack | recognition, first move, check and transfer

RECONNECT -> DISCOVER -> MAKE SENSE -> TRY -> DIAGNOSE -> FADE -> ADOPT -> TRANSFER

Diophantine Equations & Integer Restrictions

The central habit

When variables are integers, **use integrality early**. Do not solve a large real problem and hope integer answers fall out.

Use this route:

TRANSFORM -> FILTER -> FINITE CANDIDATES -> RECONSTRUCT -> CHECK

Reconnect

You already know factoring, parity, gcd, and quadratic equations. Here they become filters. A good filter removes whole families of candidates before arithmetic grows.

1. A product equation is a case generator

Suppose positive integers satisfy

$$(x-2)(y+3)=20.$$

The first useful line is not expansion. Set $x-2=d$ and $y+3=20/d$. Since the factors are integers, d must be a divisor of 20. Positivity removes some signed divisors. The factor list is not a heuristic: it is a complete parametrisation of the solutions.

Try first: Before reading further, write the signed factor-pair list for $(u)(v)=12$ and mark which pairs survive if $u>0$ and $v>3$.

A common failure is to list only positive divisors when the shifted factors may be negative. The sign conditions belong to the original variables, not automatically to the factors.

2. Difference of squares adds a parity filter

From $x^2-y^2=N$, write

$$(x-y)(x+y)=N.$$

The two factors have the same parity. For odd N , both are odd. For N divisible by 2 but not 4, there is no integer solution with both factors even. This parity check can eliminate factor pairs before reconstruction.

For $x^2-y^2=45$, factor pairs $(1,45), (3,15), (5,9)$ all have matching parity, yielding three positive solutions.

3. Coprimality changes the factor-pair problem

If $xy=N$ and $\gcd(x,y)=1$, a prime-power block of N cannot be split between both factors. Retrieve that fact from prime-exponent structure, then return here to the case problem.

Example: $N=2^4 \cdot 3^3 \cdot 7$. The blocks **4,3,7** must each go wholly to one side. That makes only 2^3 ordered allocations, instead of all divisors of 84.

This is an application of prime-exponent structure, not a second prime-factorisation lesson.

4. Bound before you enumerate

A bound turns an infinite integer problem into a finite one.

If $n(n+1)<90$ and n is positive, monotonicity gives a cheap endpoint test: $8 \cdot 9=72<90$ but $9 \cdot 10=90$. So $n \leq 8$.

For a fixed product $xy=N$ with positive integers, the smallest sum occurs at the factor pair closest to $\sqrt{N}$. A real inequality can suggest the square, but the answer is determined by actual divisor pairs. For area 96, $8 \cdot 12$ beats $6 \cdot 16$; the least perimeter is 40.

5. Exact rational approximation is an integer-gap problem

For a reduced fraction a/b near p/q ,

$$|a/b - p/q| = |qa - pb|/(qb).$$

The numerator $|qa - pb|$ is an integer. Unless the fractions are equal, it is at least 1. This converts a vague decimal-closeness question into an exact Diophantine one.

Near $3/4$ with $b \leq 15$, the best possible determinant is 1. Solving $4a - 3b = \pm 1$, the largest useful denominator is $b=15$, giving $a=11$; the sum is 26.

Contrast: Decimal testing estimates distance. The determinant certifies optimality.

6. Sum/product data reconstruct integers

If integers x,y have sum S and product P , then they are roots of

$$t^2 - St + P = 0.$$

Use this only as a representation bridge. If the target is symmetric, stop earlier. For example,

$$x^2+y^2=S^2-2P.$$

Do not solve individual roots when the requested quantity is already determined.

7. A quadratic integer root needs more than real feasibility

For $ax^2+bx+c=0$ with integer coefficients, an integer root forces the discriminant $D=b^2-4ac$ to be a nonnegative perfect square, together with the divisibility condition from the quadratic formula. This is a narrow feasibility bridge: the general discriminant theory lives elsewhere.

A useful square filter often becomes a difference of squares after completing the square. If

$n^2+6n+5=k^2$, then $(n+3)^2-k^2=4$, so $(n+3-k)(n+3+k)=4$. The fixed product creates finitely many cases.

8. One equation plus a side condition: eliminate first

With several integer variables, use the side condition early. A fixed sum may remove one variable; a gcd condition may normalize two variables; a positivity condition may bound a factor.

Historical anchor pattern: an equation in a,b,c together with $a+b+c=32$ can be rearranged to $(a-b)(c-1)=66$. The product structure and the sum cooperate; a three-dimensional brute-force search is unnecessary.

9. Completeness is part of the solution

A finite case solution is not complete until you answer:

1. Why must every solution enter one of these cases?
2. Are the cases disjoint, or have duplicates been removed?
3. Did sign, parity, gcd, and domain conditions survive reconstruction?
4. Did a one-way step create candidates that need checking?
5. Have all surviving candidates been substituted into the original condition?

For multiplicative partitions, sort factors or impose a canonical order before counting; otherwise permutations of the same factorisation masquerade as new cases.

10. Source anchors as recognition training

- A fixed-area rectangle asks for **discrete factor pairs**, not just a continuous square optimum.
- A closest fraction with bounded denominator asks for the **integer determinant $qa-pb$** .
- A quartic-looking positive-integer relation may collapse immediately modulo a shifted factor; always preserve the printed exponent.
- A quadratic Diophantine equation may become a **fixed difference of squares** after scaling.
- An equal-sum/equal-product representation becomes a **multiplicative partition plus forced ones** problem.

Closing router

Before calculating, ask:

What integer restriction collapses the search first?

Then transform only far enough to expose that restriction, enumerate a provably complete finite set, reconstruct, and check the original statement.

First-Step Reference - Diophantine Equations & Integer Restrictions

Recognition atlas

Visible clue	Structural question	First useful line		
product/shifted product equals constant	are factors integral?	set factors equal to divisor pair		
difference of squares	what parity must factor pair have?	$(x-y)(x+y)=N$		
fixed product + gcd 1	can prime-power blocks split?	allocate whole blocks		
integer maximum/minimum	can I bound before listing?	locate feasible factor pair near real optimum		
fraction near p/q	is scaled error an integer?	write $\frac{a}{b}$	$qa-pb$	$/(qb)^k$
sum and product	is target symmetric?	use S, P invariants or t^2-St+P		
quadratic must have integer root	is discriminant a square?	$D=b^2-4ac=k^2$		
many variables + fixed sum	can one variable be eliminated?	substitute side condition immediately		
finite candidates	have I proved completeness?	state case-to-solution correspondence		

Contrast strip

- Real optimum vs integer attainment: a bound suggests where to look; divisors decide what exists.
- Factorisation vs brute force: factorisation is complete only when sign/parity branches are accounted for.
- Decimal closeness vs exact rational gap: decimals estimate; $qa-pb$ certifies.
- Quadratic real root vs integer root: $D \geq 0$ is not enough; integer feasibility is stricter.
- Sum/product reconstruction vs solving variables: symmetric targets often stop before roots.
- Prime-exponent retrieval vs local teaching: use block allocation, do not rebuild divisor theory.

Final checks

Original equation; positivity/sign; parity; gcd/coprimality; denominator nonzero; integer reconstruction; duplicate removal; boundary endpoints; all one-way candidates rechecked.

Recognition and First-Line Lab

For each prompt, write only the first useful mathematical line. Do not finish the problem yet.

1. Positive integers satisfy $(x-4)(y+1)=30$.
2. Integers satisfy $x^2-y^2=72$.
3. $xy=2^5 \cdot 3^2 \cdot 7$ and $\gcd(x,y)=1$.
4. Integer rectangle area 120; perimeter is to be minimized.
5. Reduced $\frac{a}{b}$ is closest to $\frac{7}{9}$ under a denominator cap.
6. Integers x,y have sum 15 and product 44; target is x^2+y^2 .
7. A quadratic with integer coefficients is required to have an integer root.
8. Three positive integers satisfy one equation and $a+b+c=40$.
9. You have listed six factor cases. What sentence proves there cannot be a seventh kind?
10. Squaring an equation produced three integer candidates. What must happen before any is accepted?

Close decisions

A. $xy=96$, minimize $x+y$: factor pairs or all real positive pairs? Why? B. $x^2+y^2=25$: factorisation into divisor cases or geometric/square enumeration? What clue changes the route? C. $\frac{a}{b}$ nearest $\frac{3}{5}$: decimal comparison or determinant? State the exact quantity to minimize. D. $x+y=12, xy=35$: solve both variables or evaluate x^2+y^2 directly? Decide from the target.

Practice and Transfer Bank

Attempt each item before consulting the Teacher Diagnostic Key. Items 1-8 reconnect core moves, 9-16 combine filters, and 17-24 change representation or context.

1. Find all positive integer pairs x, y with $xy=36$ and $x+y$ minimal.
2. Solve positive integers $(x-2)(y+3)=20$; give the number of solutions.
3. Find positive integers x, y satisfying $xy=84$ and $\gcd(x, y)=1$; how many unordered pairs?
4. Find the greatest integer n for which $n(n+1)<90$.
5. How many integers x satisfy $x^2-5x+6=0$?
6. Find all integer x with $x^2=3x+10$; return the sum of solutions.
7. A reduced fraction a/b differs from $2/3$ by $1/30$ and $b>0$. Find $a+b$ for the smaller fraction.
8. Find the least positive denominator b for a fraction strictly between $3/7$ and $4/9$.
9. Find integer solutions to $x+y=11$, $xy=24$; give $|x-y|$.
10. If $x+y=14$ and $xy=48$ for integers x, y , find x^2+y^2 .
11. How many integer x make x^2+4x+7 a perfect square ≤ 16 ?
12. For integer n , when is n^2+n odd? Return 0 if never, 1 otherwise.
13. Find the number of positive integer pairs $x \leq y$ with $1/x + 1/y = 1/2$.
14. Positive integers x, y satisfy $x^2-y^2=45$ and $x>y$. Find the number of solutions.
15. Find the greatest possible $\gcd(x, y)$ if $x+y=72$ and xy is divisible by 100.
16. Find integer n such that n^2+6n+5 is a square and $|n| \leq 10$; how many n ?
17. A rectangle has integer sides and area 96. Find its least perimeter.
18. Find the least positive $|4a-5b|$ for integers $1 \leq b \leq 12$ with $\gcd(a, b)=1$ and $a/b \neq 5/4$.
19. How many positive integer triples $a \leq b \leq c$ have $abc=30$?
20. Find all integer x with $(x-1)(x+4)=24$; return the product of the solutions.
21. If $t^2-St+P=0$ has integer roots and $S=9, P=14$, find the larger root.
22. How many integer x make $x(12-x)$ positive and divisible by 5?
23. Positive integers x, y satisfy $xy=2^6 \cdot 3^2$ and $\gcd(x, y)=1$. How many ordered pairs?
24. Find the number of integers k with $0 \leq k \leq 20$ for which $k^2-7k+10$ is even.

Reflection after the bank

For any item that took more than one false start, record: visible clue, first line you tried, cheaper filter you missed, and the final condition that certified the answer.

Mastery Check

No method names or hints are supplied on the first attempt. Show enough work to justify completeness, not only a final number.

1. Positive integers x, y satisfy $xy=180$ and $\gcd(x, y)=1$. Minimize $x+y$.
2. Find the number of integer pairs (x, y) satisfying $(x-3)(y+2)=24$ and $x+y>0$.
3. A reduced a/b with $1 \leq b \leq 20$ is closest to $5/7$ but not equal. Find $a+b$.
4. Find all integers n with $n^2-10n+13$ a square and $0 \leq n \leq 30$; how many?
5. Integers x, y have $x+y=17$ and $xy=60$. Find $|x-y|$.
6. Positive integers $x \leq y$ satisfy $1/x + 1/y = 1/3$. Find $x+y$ for the solution with minimum y .
7. Find the least perimeter of an integer rectangle of area 210.
8. How many positive integer triples $a \leq b \leq c$ satisfy $abc=72$?
9. For integers x , how many values with $|x| \leq 20$ make x^2+x+1 divisible by 3?
10. Find the greatest $|m-n|$ over integer solutions of $m^2=4n^2-5n+16$.
11. How many reduced fractions a/b with $1 \leq b \leq 12$ satisfy $|a/b - 1/2| = 1/12$?
12. Integers x, y satisfy $x+y=20$ and $x^2+y^2=208$. Find xy .
13. Positive integers x, y satisfy $x^2-y^2=105$. Find the smallest x .
14. Find the number of integer n in $[-50, 50]$ for which $n(n+1)$ is divisible by 6.
15. If integer roots of t^2-St+P are distinct and $S=12, P=35$, find their product difference $|r_1-r_2|$.
16. How many positive integers $n < 100$ have exactly one multiplicative partition into factors > 1 after order is ignored and including n itself?